Statistical inference

5 Census and Surveys

Population vs Sample

- The population includes all objects of interest whereas the sample is only a portion of the population. Parameters are associated with populations and statistics with samples. Parameters are usually denoted using Greek letters (mu, sigma) while statistics are usually denoted using Roman letters (x, s). There are several reasons why we don't work with populations. They are usually large, and it is often impossible to get data for every object we're studying. Sampling does not usually occur without cost, and the more items surveyed, the larger the cost. We compute statistics, and use them to estimate parameters. The computation is the first part of the statistics course (Descriptive Statistics) and the estimation is the second part(Inferential Statistics)

Types of Sampling

There are five types of sampling: Random, Systematic, Convenience, Cluster, and Stratified.

- **Random sampling** is analogous to putting everyone's name into a hat and drawing out several names. Each element in the population has an equal chance of occurring. While this is the preferred way of sampling, it is often difficult to do. It requires that a complete list of every element in the population be obtained.

Computer generated lists are often used with random sampling. You can generate random numbers using the TI82 calculator.

- **Systematic sampling** is easier to do than random sampling. In systematic sampling, the list of elements is " counted off " . That is, every kth element is taken. This is similar to lining everyone up and numbering off "1,2,3,4; 1,2,3,4; etc " . When done numbering, all people numbered 4 would be used etc.
- **Cluster sampling** is accomplished by dividing the population into groups- - usually geographically. These groups are called clusters or blocks. The clusters are randomly selected, and each element in the selected clusters is used.
- **Stratified sampling** also divides the population into groups called strata. However, this time it is by some characteristic, not geographically. For instance, the population might be separated into males and females. A sample is taken from each of these strata using either random, systematic, or convenience sampling.

  In statistical surveys, when subpopulations within an overall population vary, it is advantageous to sample each subpopulation (stratum) independently. **Stratification is the process of dividing members of the population into homogenous subgroups before sampling.** The strata should be mutually exclusive: every element in the population must be assigned to only one stratum. The strata should also be collectively exhaustive: no population can be excluded. Then simple random sampling or systematic sampling is applied within each stratum. This often improves the representativeness of the sample by reducing sampling error. It

can produce a weighted mean that has less variability than the arithmetic mean of a simple random sample of the population.

In computational statistics, stratified sampling is a method of variance reduction when Monte Carlo Methods are used to estimate population statistics from a known population.

Example

Assume that we need to estimate average number of votes for each candidate in an election. Assume that country has 3 towns: Town A has 1 million factory workers, Town B has 2 million office workers and Town C has 3 million retirees. We can choose to get a random sample of size 60 over entire population but there is some chance that the random sample turns out to be not well balanced across these towns and hence is biased causing a significant error in estimation. Instead if we choose to take a random sample of 10, 20 and 30 from Town A, B and C respectively then we can produce a smaller error in estimation for the same total size of sample.

Stratified sampling strategies

Proportionate allocation uses a sampling fraction in each of the strata that is proportional to that of the total population. For instance, if the population X consists of m in the male stratum and f in the female stratum (where m + f=X), then the relative size of the two samples (x1=m/X males, x2= f/X females) should reflect this proportion.

Optimum allocation (or Disproportionate allocation) – Each stratum is proportionate to the standard deviation of the distribution of the variable. Larger samples are taken in the strata with the greatest variability to generate the least possible sampling variance.

Stratified sampling ensures that at least one observation is picked from each of the strata, even if probability of it being selected is close to 0. Hence the statistical properties of the population may not be preserved if there are thin strata. A rule of thumb that is used to ensure this is that the population should consist of no more than six strata, but depending on special cases the rule can change – for example if there are 100 strata each with 1 million observations, it is perfectly fine to do a 10% stratified sampling on them.

A real-world example of using stratified sampling would be for a political survey. If the respondents needed to reflect the diversity of the population, the researcher would specifically seek to include participants of various minority groups such as race or religion, based on their proportionality to the total population as mentioned above. A stratified survey could thus claim to be more representative of the population than a survey of simple random sampling or systematic sampling.

Advantages

The reasons to use stratified sampling rather than simple random sampling include:

- If measurements within strata have less standard deviation, the stratification gives smaller error in estimation.
- For many applications, measurements become more manageable and/or cheaper when population is grouped into strata.
- It is often desirable to have estimates of population parameters for groups within the population.
- If population density varies greatly within a region, stratified sampling will ensure that estimates can be made with equal accuracy in different parts of the region, and that comparisons of sub-regions can be made with equal statistical power. For example, in Ontario a survey taken throughout the province might use a larger sampling fraction in the less populated north, since the disparity in population between north and south is so great that a sampling fraction based on the provincial sample as a whole might result in the collection of only a handful of data from the north.

Randomized stratification can also be used to improve population representativeness in a study.

Disadvantages

Stratified sampling is not useful when the population cannot be exhaustively partitioned into disjoint subgroups. It would be a misapplication of the technique to make subgroups' sample sizes

to subgroup sizes proportional to the amount of data available from the subgroups, rather than scaling sample sizes to subgroup sizes (or to their variances, if known to vary significantly e.g. by means of an F Test) . Data (at the same time) to the subgroups' sizes within the total population. For an efficient way to partition sampling resources among groups that vary in their means, their variances, and their costs, see "optimum allocation". The problem of stratified sampling in the case of unknown class priors (ratio of subpopulations in the entire population can have deleterious effect on the performance of any analysis on the dataset, e.g. classification. In that regard, mini-max sampling ratio can be used to make the dataset robust with respect to uncertainty in the underlying data generating process.